

V. Write short answers:

1. Differentiate between if and switch statement. [ICSE 2006]
2. Explain with the help of an example, the purpose of default in a switch statement. [ICSE 2005]
3. What are the three ways to take the values in a Java Programming?
4. Write down the syntax with reference to Java Programming:
 - (a) to accept an integral value from the user through console
 - (b) to accept a fractional number from the user through console
 - (c) if-else statement
 - (d) switch case
 - (e) if-else if construct with an example. [ICSE 2007]

5. What is the use of the keyword import? [ICSE 2010]
6. What is a Package? Give an example. [ICSE 2006]
7. Distinguish between:
 - (a) Testing and Debugging
 - (b) Syntax error and Logical error

8. What are the different types of errors that take place during the execution of a program?
9. Write one word answer:

- (i) A method that converts a string to a primitive integer data type
- (ii) The default initial value of a boolean variable data type [ICSE 2009]
10. What is a compound statement? Give an example. [ICSE 2005]
11. What do you understand by 'Fall through'? Give an example.

VI. Java Programming:

1. Mr. Shyam Lal Agarwal invests certain sum at 5% per annum compound interest for three years. Write a program in Java to calculate:
 - (a) the interest for the first year
 - (b) the interest for the second year
 - (c) the amount after three years.
 Take sum as an input from the user.
2. A shopkeeper offers 10% discount on the printed price of a Digital Camera. However, a customer has to pay 6% sales tax on the remaining amount. Write a program in Java to calculate the amount to be paid by the customer, taking printed price as an input.
3. A man invests certain sum of money at 5% compound interest for the first year, 8% for the second year and 10% for the third year. Write a program in Java to calculate the amount after three years, taking the sum as an input.

[Hint: $A = P(1 + R_1/100)(1 + R_2/100)(1 + R_3/100)$]

4. A shopkeeper offers 30% discount on purchasing articles whereas the other shopkeeper offers two successive discount 20% and 10% for purchasing the same articles. Write a program in Java to compute and display which is better offer for a customer. Take the price of an article as an input. **HW**

5. Mr. Guha wishes to accumulate 3000 shares of a company. However, already he has some shares of that company valuing ₹ 10 (nominal value) which yield 10% dividend per annum and received ₹ 2000 as dividend at the end of the year. Write a program in Java to calculate the number of shares he has and how many more shares to be purchased to make his target.

[Hint: $\text{No. of share} = \frac{\text{Annual dividend} \times 100}{\text{Nominal value} \times \text{div}\%}$] **HW**

6. Write a program in Java to calculate the value of $\sin 30^\circ$ (degree), $\cos 30^\circ$ (degree) and $\tan 30^\circ$ (degree). Now, display the lowest value of the trigonometrical ratios.
7. Write a program in Java to accept the cost price and the selling price of an article and calculate either profit percent or loss percent. **HW**
8. Write a program to input three sides of a triangle and check whether the triangle is possible or not. If possible then display whether it is an Equilateral, Iso-sceles or a Scalene Triangle otherwise, display 'Triangle not possible'. **HW**

9. 'Atul Transport' charges for carrying parcels as per the city and the weight. The tariff is given below:

City	Weight	Charges
Kolkata	Up to 100 kg	₹ 45/kg
	More than 100 kg	₹ 75/kg
Mumbai	Up to 100 kg	₹ 65/kg
	More than 100 kg	₹ 95/kg
Chennai	Up to 100 kg	₹ 75/kg
	More than 100 kg	₹ 115/kg
Delhi	Up to 100 kg	₹ 90/kg
	More than 100 kg	₹ 125/kg

Write a program to input city name 'K' for Kolkata, 'M' for Mumbai, 'C' for Chennai, 'D' for Delhi and weight of the parcel. Calculate and display the charges to be paid by a customer for booking.

10. Write a program in Java to compute charges for sending parcels when the charges are as follows:

For the first 1 kg: ₹ 15.00

HW

For additional weight, for every 500 gm or fraction thereof: ₹ 8.00 [ICSE 1999]

11. To foster a sense of water conservation, the water department has an annual water conservation tax policy. The rates are based on the water consumption of the consumer. The tax rates are as follows:

Water consumed (in Gallons)	Tax Rate in ₹/100 gallons
Up to 45	No Tax
More than 45 but 75 or less	475.00
More than 75 but 125 or less	750.00
More than 125 but 200 or less	1225.00
More than 200 but 350 or less	1650.00
More than 350	2000.00

Write a program in Java to compute and display an annual water tax chart based on the input value of water consumption. [ICSE 2003]

12. Write a program to input the code of a particular item, the quantity purchased and the rate. Then ask if any more items are to be purchased. If not, then calculate the total purchase price and print it along with the gift to be presented. The gifts to the customers are given in the following manner:

Amount of Purchase (₹)	Gift
100 and above but less than 500	A key ring
500 and above but less than 1000	A leather purse
1000 and above	A pocket calculator

Before ending the program a "Thank you" statement should be printed to all the customers. [ICSE 1997]

13. A library charges a fine for books returned late as given below:

First five days	40 paise per day	HW
Six to ten days	65 paise per day	
Above ten days	80 paise per day	

Design a program in JAVA (with relevant supporting material) to calculate the fine assuming that a book is returned N days late. [ICSE 1996]

14. Mr. A. P. Singh is a software engineer. He pays annual income tax as per the given table:

Annual Salary	Rate of Income Tax
Up to ₹ 1,00,000	No Tax
₹ 1,00,001 to ₹ 1,50,000	10% of the amount exceeding ₹ 1,00,000
₹ 1,50,001 to ₹ 2,50,000	₹ 5000 + 20% of the amount exceeding ₹ 1,50,000
Above ₹ 2,50,000	₹ 25,000 + 30% of the amount exceeding ₹ 2,50,000

Write a program in Java to compute the income tax to be paid by him. HW

15. St. Xavier School displays a notice on the school notice board regarding admission in Std. XI for choosing different streams, according to marks obtained in English, Maths and Science in ICSE Examinations.

Marks obtained in diff. subjects	Stream	HW
Eng., Maths and Science $\geq 80\%$	Pure Science	
Eng. and Science $\geq 80\%$, Maths $\geq 60\%$	Bio. Science	
Eng., Maths and Science $\geq 60\%$	Commerce	

Print the appropriate stream allotted to the candidate. Write a program in Java to accept marks in English, Maths and Science from the console.

16. Enter the record of the employees of a company as employee number, age and basic salary. Write a program to calculate monthly gross salary and net salary of each employee as follows:

Dearness Allowance	Nature (50% of Basic Salary) HW
House Rent Allowance	40% of Basic Salary
Provident Fund	12% of Basic Salary
Employee Provident Fund	2% of Basic Salary

Gross Salary = Basic + DA + HRA

Net Salary = Gross Salary - (PF + EPF)

The program further checks whether an employee is an Income Tax payer or not, as under:

Category	Annual Gross Salary
Male (M)	Over ₹ to ₹ 1,50,000 Income Tax Payer
Female (F)	Up to ₹ 2,00,000 Not an Income Tax Payer

Enter 'M' or 'F' to check the category to decide whether an employee has to pay Income Tax or not. Print employee number, gross salary, net salary and also an 'Income Tax' payer or not.

17. 'Bank of India' announces new rates for Term Deposit Schemes for their customers and Senior Citizens as given below:

HW

Rate of Interest (General) Rate of Interest (Senior Citizen)

Term	Rate of Interest (General)	Rate of Interest (Senior Citizen)
Up to 1 year	7.5%	8.0%
Up to 2 years	8.5%	9.0%
Up to 3 years	9.5%	10.0%
More than 3 years	10.0%	11.0%

The senior citizen is applicable to those customers, whose age is 60 years or more. Write a program to accept the sum (p) in term deposit scheme, age of the customer and the term. The program displays the information in the given format:

Amount Deposited	Term	Age	Interest earned	Amount Paid
xxx	xxx	xxx	xxx	xxx

18. A special two-digit number is such that when the sum of its digits is added to the product of its digits, the result is equal to the original two-digit number.

Example: Consider the number 59.

$$\text{Sum of digits} = 5 + 9 = 14$$

$$\text{Product of digits} = 5 * 9 = 45$$

$$\text{Total of the sum of digits and product of digits} = 14 + 45 = 59$$

HW

Write a program to accept a two-digit number. Add the sum of its digits to the product of its digits. If the value is equal to the number input, output the message "Special 2 - digit number" otherwise, output the message "Not a special two-digit number".

[ICSE 2014]

Menu Driven/Switch Case programs

19. Using Switch Case, write a program that allows to accept any values from 1 to 7 and displays the weekdays corresponding to the number entered from the keyboard.

[ICSE 2002]

20. Write a menu driven program to find the sum, difference and product of two numbers. Perform the tasks accordingly;

Enter '+' to find the sum, '-' to find the difference and '*' to find the product of two numbers.

21. Using a switch case statement, write a menu driven program to convert a given temperature from Fahrenheit to Celsius and vice versa. For an incorrect choice, an appropriate error message should be displayed.

[ICSE 2007]

[Hint: $C = \frac{5}{9} (F - 32)$ and $F = 1.8 * C + 32$]

HW.

22. Write a menu driven program to calculate:

1. Area of a circle = πr^2 ($\pi = 22/7$)

2. Area of a square = side*side

3. Area of a rectangle = length*breadth

HW.

Enter 'c' to calculate area of circle, 's' to calculate area of square and 'r' to calculate area of rectangle.

23. Write a menu driven program that outputs the results of the following evaluations based on the number entered by the user:

(a) Natural logarithm of the number

(b) Absolute value of the number

(c) Square root of the number

(d) Random numbers between 0 and 1.

[ICSE 2006]

HW.

24. Write a program using switch case to find the volume of a cube, a sphere and a cuboid. For an incorrect choice, an appropriate error message should be displayed.
- Volume of a cube = $s * s * s$
 - Volume of a sphere = $\frac{4}{3} * \pi * r * r * r$ **HW .**
 - Volume of a cuboid = $l * b * h$
25. A Promoter cum Developer announces a special bonanza for early booking of the flats for their customers as per the tariff given below: **HW .**

Category	Discount on the price	Discount on Development Charge
Ground floor	10%	8%
First floor	2.0%	1.0%
Second floor	5%	5%
Third floor	7.5%	10%

Write a menu driven program to input price and the category '0' (zero) for ground floor, '1' for first floor, '2' for second floor and '3' for third floor). Calculate and display the total discount, price of the flat after getting discount.

26. A company announces revised Dearness Allowance (DA) and Special Allowances (SA) for their employees as per the tariff given below:

Basic	Dearness Allowance (DA)	Special Allowance (SA)
Up to ₹10,000	10%	5%
₹10,001 – ₹20,000	12%	8%
₹20,001 – ₹30,000	15%	10%
₹30,001 and above	20%	12%

Write a program to accept Basic Salary (BS) of an employee. Calculate and display gross salary.

Gross Salary = Basic + Dearness Allowance + Special Allowance **HW .**

Print the information in the given format:

Basic	DA	Spl. Allowance	Gross Salary
xxx	xxx	xxx	xxx

27. You want to buy an old car from 'Sale and Purchase', where you can get a car in depreciated price. The depreciation value of a car is calculated on its showroom price and the number of years it has been used. The depreciated value of a car is calculated as per the tariff given below:

No. of years used	Rate of depreciation
1	10%
2	20%
3	30%
4	50%
Above 4 years	60%

HW .

Write a menu driven program to input showroom price and the number of years used ('1' for one year old, '2' for two years old) and so on. Calculate the depreciated value. Display the original price of car, depreciated value and the amount paid for the car.

28. An electronics shop has announced the following seasonal discounts on purchase of certain items.

Category	Discount on Laptop	Discount on Desktop PC
Up to ₹ 25,000	0.0%	5.0%
₹ 25,001 – ₹ 57,000	5.0%	7.5%
₹ 57,001 – ₹ 1,00,000	7.5%	10.0%
More than ₹ 1,00,000	10.0%	15.0%

Write a program based on above criteria, to input name, address, amount of purchase and the type of purchase ('L' for Laptop and 'D' for Desktop). Compute and print the net amount to be paid by a customer with his name and address.

[Hint: discount = Discount rate/100)*Amount of purchase

Net amount = Amount of purchase – Discount]

[ICSE 2009]

HW